

Curriculum Vitae

Personal information**Name:** Melian, Carlos**Researcher ID:** C-2146-2008 **ORCID:** 0000-0003-3974-6515 **SCOPUS:** 55220360000**Date of birth:** 23/08/1969**Nationality:** Spanish**URL for web site:** <http://www.eawag.ch/en/aboutus/portrait/organisation/staff/profile/carlos-melian/show/>**Education**

2005 PhD in Biology and Environmental Sciences.
 Department of Ecology, University of Alcalá, Madrid, and Biological Doñana Station,
 Seville, Spain. Supervisor: Prof. Jordi Bascompte.

Professional Experience**2015-Present** Lecturer, University of Bern, Switzerland.**2010-Present** Group leader (Tenured since end 2015),
 Swiss Federal Institute of Aquatic Science and Technology, Switzerland.**Fellowships****2008-2010** **Postdoctoral Fellow**, Microsoft Research Ltd. UK, (Unrestricted Gift).**2005-2007** **Postdoctoral Fellow**, National Center of Ecological Analysis and Synthesis,
 University of California Santa Barbara, USA.**2001-2004** **PhD. Fellowship**, Spanish Ministry of Science and Technology, Spain**1999** **Student research fellowship**, Ministry of Foreign Affairs. Spanish Agency for
 International Cooperation, Spain and University of Granma, Bayamo, Cuba.
 Title: Linking Agroecology, Conservation and Theoretical Ecology.**1997** **Student research fellowship**, Department of Ecology, Alcala University, Spain
 Title: Complex Systems: Theory and its Application to Ecology.**Organization of international scientific meetings****2016** International exploratory workshop “Interactions on evolutionary trees”,
 Swiss National Science Foundation, Switzerland.**2008** Workshop “Theoretical ecology: Big questions and challenges”. National Center for
 Ecological Analysis and Synthesis (NCEAS). Univ. California Santa Barbara, USA.**Supervision and co-supervision of Masters, PhD students, and Postdocs****PhD students****2012-2015** (PhD) student Simone Fontana with Dr. Francesco Pomati
 Swiss National Science Foundation Grant 31003A-144162. Switzerland.**2010-2014** (PhD) PhD. Student Francisco Baldo with Dr. Pilar Drake
 PhD. Fellowship, Spanish Ministry of Science and Technology, Spain.**Postdocs****2017** Gian Marco Palamara (Lab discretionary funding)**2015** Ayana Martins (BEPE fellowship, Brazil)**2013-2015** Charles N. de Santana (SNSF 31003A_144053)**2013-2014** Jan Klecka (SCIEX 12.327)**2013-2014** Gregor Kalinkat (German Academic Exchange Service, DAAD)**2012** Josephine Rodriguez (Lab discretionary funding)**Editorial and Boards****2011-Present** Subject editor *Oikos***2018** Recommender for PCI Ecology**Teaching activities****2013-present** “Introduction to literature analysis”

Institute of Ecology and Evolution, Uni. Bern, Switzerland.

2013 Mini-course “Speciation models” to undergraduates and PhD students. Eawag,
 Kastanienbaum, Switzerland.**2012** “Evolutionary biodiversity dynamics” with Prof. Ole Seehausen. Institute of Ecology
 and Evolution, U. Bern, Switzerland.**2007-2009** “Kids Do Ecology” program. National Center for Ecological Analysis and
 Synthesis (NCEAS). Univ. California Santa Barbara, USA.

Invited talks (top 5 highlighted)

- 2016** **MEDECOS conference** Seville, Spain
 “Human driven scenarios for evolutionary and ecological changes”
 Keynote speaker symposia
 Title: Interdependent Networks in Ecology and Evolution
- 2015** **Mathematical models in Ecology and Evolution conference**, Paris, France
 Title: Eco-evolutionary diversification dynamics.
- 2013** **INTECOL meeting**, London, UK
 Title: Connecting diversification and biodiversity dynamics across spatial scales, INTECOL meeting, London, August, UK.
- 2012** **Web of life conference**, University of Montpellier, France.
 Title: Eco-evolutionary dynamics of individual-based networks,
- 2011** **ESA, Ecological Society of America Meeting**, Austin, Texas, USA.
 Title: Testing adaptive and non-adaptive radiations and biodiversity patterns in supernetworks,

Publication summary

I have 29 publications in international peer reviewed journals, h-index: 18, i-10 index: 25, total citations: 3320 (Google scholar May 29, 2018). I have 17 papers as first/senior author with a mean 5YIF equals to 8.924 of which 13 are without my PhD supervisor with a mean 5YIF equals to 7.3611.

Grant summary

- 2016** **Melián, C. J.**, Matthews, B., Seehausen, O., and Harmon, L. J.
 Granted: Swiss National Science Foundation, International exploratory workshops
 Title: Interactions on Trees.
 Period: 1 Week, 21,000 CHF.
- 2014-2015** Kalinkat, G., and **Melián, C. J.** (PI)
 Granted: German Academic Exchange Service (DAAD). Germany
 Title: Analysing the interplay between allometric constraints and intraspecific trait variation to predict food web dynamics.
 Period: 6 Months, 17,694€.
- 2014-2015** **Melián, C. J.** (PI) and Klecka, J.
 Granted: Swiss National Science Foundation, International Coop. Switzerland.
 Title: Modeling the effects of predator phenotype on food web structure.
 Period: 3 Months, 9,510 CHF.
- 2012-2015** **Melián, C. J.** (PI)
 Granted: Swiss National Science Foundation, Division III. Switzerland.
 Title: A theory for next-generation food web data.
 Period: 1 and a half Years (Postdoc), 161,000 CHF.
- 2012-2014** **Melián, C. J.** (PI), Klecka, J., Křivan, V., and Altermatt, F.
 Granted: Sciex, Switzerland.
 Title: Food webs in space: integrating metacommunity and food web research.
 Period: 1 Year, 92,900 CHF

Prices and awards

- 2008-2010** **Postdoctoral Fellow** granted by Microsoft Research Ltd. (Unrestricted Gift).
Place: NCEAS, University of California, Santa Barbara, USA.
Title: Unifying Theories of Molecular, Community and Network Evolution.
Period: 2 Years, 200,000\$.
Project page: <https://www.nceas.ucsb.edu/projects/12306>

Peer-reviewed articles (five last years 2014-2018)

1. **Melián, C. J.**, Matthews, B., de Andreazzi, C. S., Rodriguez, J. P., Harmon, L., and Fortuna, M. A. (2018). Deciphering the interdependence between ecological and evolutionary networks. *Trends in Ecology and Evolution* doi.org/10.1016/j.tree.2018.04.009. [IF:15.268]
2. de Andreazzi, C. S., *Guimaraes, P., and ***Melián, C. J.** (2018). Eco-evolutionary feedbacks promote fluctuating selection and long-term stability of antagonistic networks. *Proceedings of the Royal Society B*, 285:20172596. *Last co-authors. [IF:4.94]
3. Pellissier, L., Albouy, C., Bascompte, J., Farwig, N., Graham, C., Loreau, M., Maglianesi, M. A., **Melián, C. J.**, Pitteloud, C., Roslin, T., Rohr, R., Saavedra, S., Thuiller, W., Woodward, G., Zimmermann, N. E., and Gravel, D. (2017). Comparing species interaction networks along environmental gradients, *Biological Reviews*, 93:785-800. [IF:11.615]
4. De Laender, F., Rohr, J. R., Ashauer, R., Baird, D. J., Berger, U., Eisenhauer, N., Grimm, V., Hommen, U., Maltby, L., **Melián, C. J.**, Pomati, F., Roessink, I., Radchuk, V., and Van den Brink, P. J. (2016). Reintroducing environmental change drivers in biodiversity–ecosystem functioning research. *Trends in Ecology and Evolution*, 31:905-915. [IF:15.268]
5. Leprieur, F., Descombes, P., Gaboriau, T., Cowman, P. F., Parravicini, V., Kulbicki, M., **Melián, C. J.**, de Santana, C. N. Heine, C., Mouillot, D., Bellwood, D. R., and Pellissier, L. (2016). Plate tectonics drive tropical reef biodiversity dynamics. *Nature Comm.*, 7:11461. [IF:12.124]
6. *Fronhofer, E., *Klecka, J., **Melián, C. J.**, and Altermatt, F. (2015). Condition-dependent movement and dispersal in experimental metacommunities. *Ecology Letters*, 18:954-963. *First co-authors. [IF:9.449]
7. Hines, J., van der Putten, W. H., De Deyn, C. B., Wagg, C., Voigt, W., Mulder, C., Weisser, W. W., Engel, J., **Melián, C. J.**, Scheu, S., Birkhofer, K., Ebeling, A., Scherber, C., and Eisenhauer, N. (2015). Towards an integration of biodiversity–ecosystem functioning and food web theory to evaluate relationships between multiple ecosystem services. *Advances in Ecological*

Research 53:161-199. [IF:5.056]

8. **Melián, C. J.**, Seehausen, O., Fortuna, M. A., Eguíluz, V., and Deiner, K. (2015). Diversification and biodiversity dynamics of hot and cold spots. *Ecography*, 38:393-401. [IF:4.902]
9. Sanz-Aguilar, A., Jovani, R., **Melián, C. J.**, Pradel, R., and Tella, J. L. (2015). Multievent capture-recapture analysis reveals individual foraging specialisation in a generalist species. *Ecology*, 96:1650–1660. [IF:4.809]
10. ***Melián, C. J.**, *Křivan, V., Altermatt, F., Stary, P., Pellissier, L., and De Laender, F. (2015). Dispersal dynamics in food webs. *The American Naturalist*, 185:157-168. *First co-authors. [IF: 4.167]
11. **Melián, C. J.**, Baldó, F., Matthews, B., Vilas, C., González-Ortegón, E., Drake, P., and Williams, R. J. (2014). Intraspecific trait variation and diversity in food webs. *Advances in Ecological Research*, 50:207-241. [IF:5.056]
12. Moya-Laraño, J., Bilbao-Castro, J. R., Barrionuevo, G., Ruiz-Lupión, D., Casado, L. G., Montserrat, M., **Melián, C. J.**, and Magalhães, S. (2014). Eco-evolutionary spatial dynamics: rapid evolution and isolation explain food web persistence. *Advances in Ecological Research*, 50:75-143. [IF:5.056]
13. *De Laender, F., ***Melián, C. J.**, Bindler, R., Van den Brink, P., Daam, M., Roussel, H., Juselius, J., Verschuren, D., and Janssen, C. (2014). The Contribution of intra- and interspecific tolerance variability to biodiversity changes along toxicity gradients. *Ecology Letters*, 17:72–81. *First co-authors. [IF:9.449]